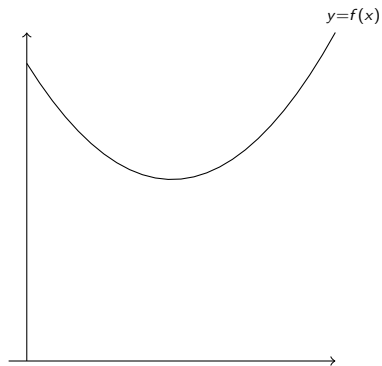
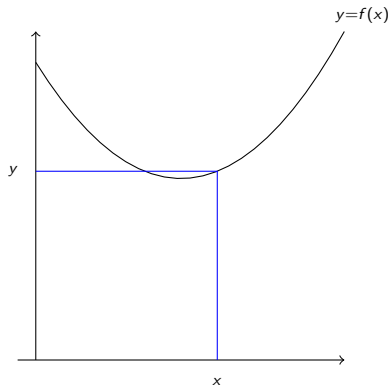


# Slopes

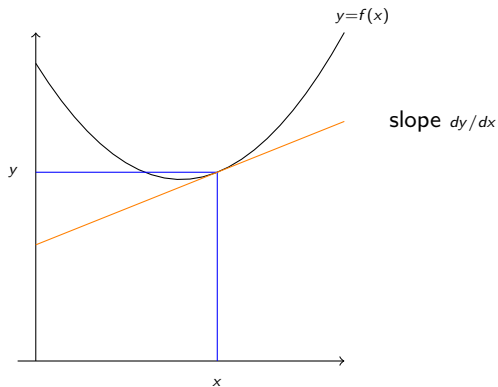


# Slopes



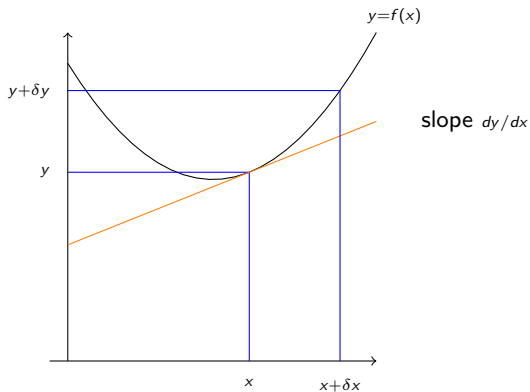
Consider variables  $x$  and  $y$  related by  $y = f(x)$ .

# Slopes



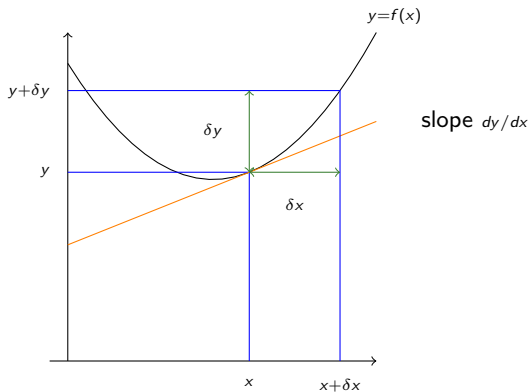
$dy/dx$  is the slope of the tangent line to the graph.

# Slopes



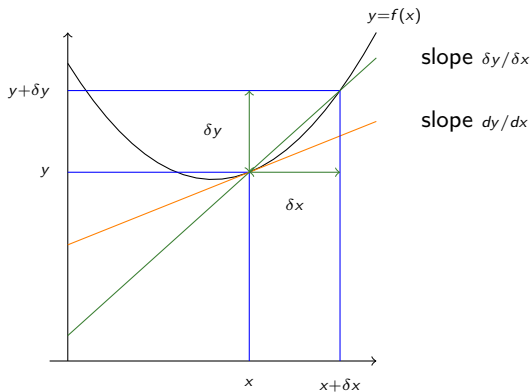
If  $x$  changes by a small amount  $\delta x$ , then  $y$  will change by a small amount  $\delta y$ .

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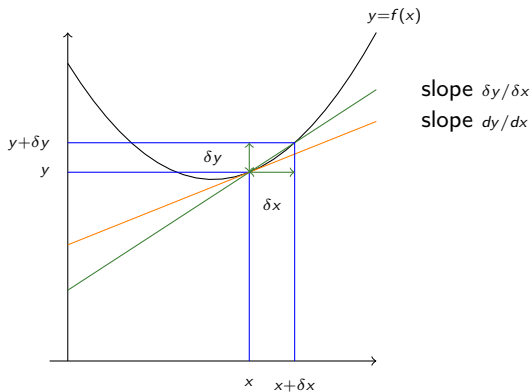
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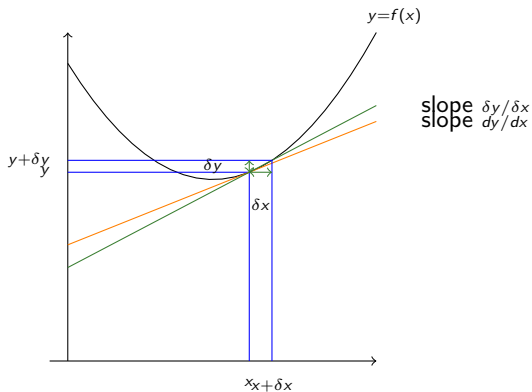
The ratio  $\delta y/\delta x$  is the slope of a chord cutting across the graph.

# Slopes



The slope of the chord changes slightly as  $\delta x$  decreases.

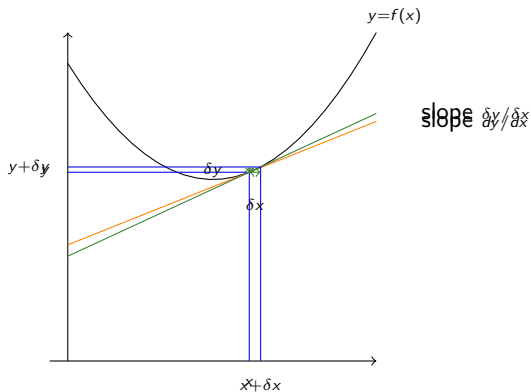
# Slopes



As  $\delta x$  approaches zero, the chord approaches the tangent, and  $\delta y / \delta x$  approaches  $dy / dx$ .

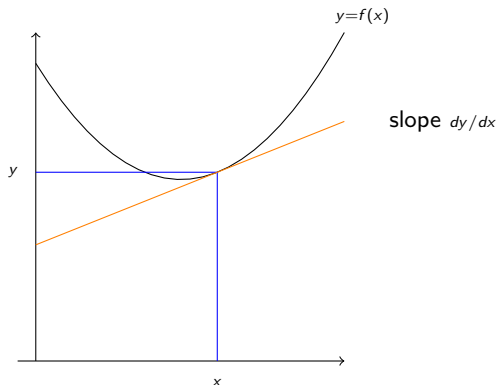


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► Conclusion:  $\exp'(x) = \exp(x)$ .